ONE-SLIDE PROPOSAL · RIDES · DROP-OFF ACCURACY

Pathao drops commuters on the wrong road. Smart Drop-off fixes the last 100 metres.

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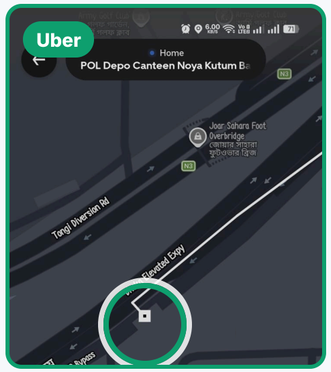
Same office, same 9:25 a.m. request: **Uber's route ends at the canteen**. **Pathao's ends across a divided highway**, on about half of my weekday trips.

1 THE MOMENT WE'RE LOSING

POL Depo Canteen Noya Kutum Bari, Dhaka



Pin on the Elevated Expressway. Route ends on the highway, opposite the canteen.

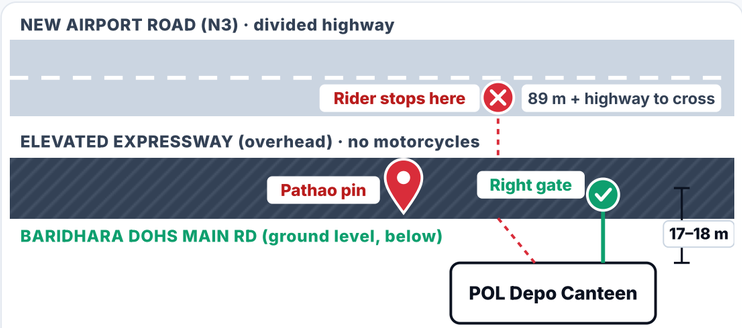


Route turns off the highway and ends at the canteen.

~1 in 2 of my office trips end on the highway. I cross a divided road on foot, or I switch to Uber.

2 WHY IT HAPPENS

Pathao snaps the pin to a road bikes can't use



- The expressway deck sits **almost exactly above** the canteen's road: 18 m vs 17 m away. On a flat map they are one line, and Pathao's pin lands on the **expressway**.
- Bikes can't use it, so navigation sends the rider to the **nearest legal road, N3**. The rider stops there, across the highway.
- The right road costs the rider only **+52 m and ~40 s**: one left turn off N3. Miss that turn and it's a 2.5 km detour.

Hypothesis built from OpenStreetMap road data and OSRM routes. Confirm with Pathao's geocoder and trip logs.

3 THE FIX: SMART DROP-OFF

Better GPS logic on the backend, plus coordination between passenger and rider

A — Access-aware snapping BACKEND
Never snap a drop-off to a road where bikes can't stop: elevated roads, ramps, motorways, no-motorcycle roads. Prefer the road the venue's entrance faces.

B — Confirm your gate, once PASSENGER
If the pin is on a no-stop road, one card asks for the right entrance and shows the walk. It's saved to "Office", so it's never asked again.

C — Gate shown to the rider RIDER
The rider sees the gate pin, the approach road and a note in Bangla before accepting. If they end the trip far from the gate, the passenger is asked to confirm.

① Backend fix → ② Gate card + rider pin → ③ Learn from trips

WHY MANAGEMENT SHOULD APPROVE IT

- **It protects commuters, our most frequent riders.** A daily route that fails half the time teaches them to open Uber first. At 2 trips a day, that's ~22 bad drop-offs a month for one person.
- **It's small and low-risk.** It uses road attributes and trip GPS we already have, and step ① ships silently on the server.
- **It works everywhere, not just here.** Every venue next to a flyover or expressway has the same bug.

HOW WE'LL KNOW IT WORKED

- **Primary:** drop-off error, the distance from where the trip ended to the confirmed destination. Target: median under 25 m.
- **Also:** "wrong drop-off" tickets, rider–passenger calls in the last 2 minutes, and 30-day commuter retention.
- **Guardrails:** rider trip time, fare and acceptance rate. 4-week A/B test in Dhaka.

OPEN QUESTIONS BEFORE THIS SHIPS

- How many Dhaka destinations snap to no-stop roads? One trip-log query sizes it.
- Does our map provider give us bridge, layer and access tags, or must we derive them?
- Can riders enter every gate (cantonment and DOHS rules)? Who checks gates that passengers suggest?
- Does the rider need an incentive for the extra turn-in?

Screenshots: my own trips, same destination, same time (9:25). Distances: OpenStreetMap (ODbL). Routes: OSRM. "~1 in 2" is my own commute log, not company data.

Clickable prototype: SafwatSameer.github.io/smart-dropoff/prototype